B.Sc. 6th Semester (Honours) Examination, 2025

COMPUTER SCIENCE

Course ID: 61516 **Course Code: SH/CSC/603/DSE-3**

Information Security

(CBCS OLD 2017-18)

Time: 1 Hr 15 Min **Full Marks: 25**

The figures in the right-hand margin indicate marks.
Candidates are required to give their answer in their own words as far as practicable.

Answer the questions unit wise as instructed

UNIT- I

1. Answer any five of the following questions: $5 \times 1 = 5$

a) Write a difference between cryptography and steganography.
b) What is meant by decryption?
c) Write a difference between virus and worm.
d) What do you mean by hacking?
e) Differentiate between symmetric key cipher and asymmetric key cipher.
f) What is block cipher?
g) What is replay attack?
h) Define spyware.

UNIT - II

2. Answer any two of the following questions: $2 \times 5 = 10$

a) Explain goals of using Information Security. What is MAC? 4+1

b) What is active cyber attack and what is passive cyber attack? Give example of each. What is watermarking? 2+2+1

c) Explain the DES round key generation process in brief.

[Turn Over]

d) Write properties of a good hash function. What is avalanche effect? Differentiate between attack and threat. 2+1+2

UNIT - III

3. Answer any one of the following questions: $1 \times 10 = 10$

a) Explain the Random Oracle model briefly. What is pre image attack, 2^{nd} pre image attack and collision attack? 4+6

b) Explain the features of a firewall. What is packet filtering firewall? Briefly explain possible attacks and countermeasures on a packet filtering firewall. 3+1+6

B.Sc. 6th Semester (Honours) Examination, 2025

COMPUTER SCIENCE

Course ID: 61516 **Course Code: SH/CSC/603/DSE-3**

Cryptographic Applications

(CBCS NEW 2022-23)

Time: 1 Hr 15 Min **Full Marks: 25**

The figures in the right-hand margin indicate marks.
Candidates are required to give their answer in their own words as far as practicable.

Answer the questions unit wise as instructed

UNIT- I

1. Answer any five of the following questions: $5 \times 1 = 5$

a) Give the list of essential security services.
b) Define the type of security attack in the case when a student breaks into a professor's office to obtain a copy of the next day's test.
c) What is Steganography?
d) What is meant by DOS attack?
e) Differentiate between symmetric key cipher and asymmetric key cipher.
f) What is traffic padding?
g) What is replay attack?
h) Define spyware.

UNIT - II

2. Answer any two of the following questions: $2 \times 5 = 10$

a) Write differences between worms and virus. What is adware?

3+2=5

[Turn Over]

b) What is active cyber attack and what is passive cyber attack? Give example of each. What is cipher block chaining? 2+2+1

c) Explain the DES function in brief.

d) Explain the components of network security model.

UNIT - III

3. Answer any one of the following questions: $1 \times 10 = 10$

a) What are the requirements of cryptographic hash function? Distinguish between diffusion and confusion. 6+4

b) Write RSA key generation algorithm. Explain any two potential attacks on RSA. 5+5

Bankura Sammilani College

Dept. of Computer Science

Internal Assessment Examination-2025

Sem-6 **Paper: Information Security**

F.M-10 **Time: 30 Minutes**

Answer any five questions.

1. Name the three security goals and explain any one.
2. What is replay attack?
3. Name the attacks threatening integrity?
4. What is denial of service attack?
5. What is digital signature?
6. What is steganography?
7. Differentiate between confusion and diffusion.
8. Differentiate between symmetric key cipher and asymmetric key cipher.

B.Sc. 6th Semester (Honours) Examination, 2025

COMPUTER SCIENCE

Course ID: 61526 **Course Code: SH/CSC/603/DSE-3**

Cryptographic Application (Practical)

(CBCS NEW 2022-23)

Time: 2 Hours **Full Marks: 15**

The figures in the right-hand margin indicate marks.
Candidates are required to give their answer in their own words as far as practicable.

(LNB + Viva = 05 Marks, Experiment = 10 Marks)

Perform any one experiment from the following:

1. Implement encryption and decryption of rail-fence cipher.
2. Design and implement a product cipher using substitution and transposition ciphers.
3. Perform encryption and decryption of affine cipher.
4. Implement Diffie Hellman Key exchange algorithm.
5. Implement RSA public key cryptosystem.
6. Use Password cracking tool 'John the Ripper' to verify the strength of a password of your own choice.
7. WAP to implement each round of Feistel Cipher.

helloworld

```
h   o   l
 e l w r
  l   o
```

[Turn Over]

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B.SC (HONOURS) SIXTH SEMESTER MOCK TEST - 2025

Subject: Computer Science

Course ID: 61516 Course Title: Information Security

Full Marks: 25 Time: 1 Hr 15 Min.

1. Answer *any five* of the following questions: (5x1=5)

a) What is steganography?

b) What is meant by ciphertext?

c) Write a difference between virus and worm.

d) What is watermarking?

e) Define digital signature.

f) What is brute force attack?

g) What is residual risk?

h) What is proxy firewall?

2. Answer *any two* of the following questions: (5x2=10)

a) Explain goals of using Information Security. What is MAC? 4+1

b) The encryption key in a transposition cipher is (3, 2, 6, 1, 5, 4). Find the decryption key. With these keys show the encryption and decryption process for the message "information secrecy" [excluding blank space] 1+2+2

c) Write two differences between symmetric key and asymmetric key encryption algorithm. With suitable diagram explain the key generation process of DES encryption algorithm (no table is required). 2+3

d) Write properties of a good hash function. What is avalanche effect? Differentiate between attack and threat. 2+1+2

3. Answer *any one* of the following questions: (10x1=10)

a) Explain the Random Oracle model briefly. What is pre image attack, 2nd pre image attack and collision attack? 4+6

b) Explain the RSA algorithm briefly. Describe the DES key generation process briefly. 5+5